

Sumaiya Akter

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RESEARCH INTEREST

Product Design, Advanced Manufacturing, Supply Chain, Ergonomics

EDUCATION AND CERTIFICATIONS

Bangladesh University of Engineering and Technology (BUET)

Dhaka

Bachelor of Science in Industrial and Production Engineering

February 2017 - May 2022

- **CGPA: 3.36/4.00**

RESEARCH EXPERIENCE

Impacts of risk factors on the supply chain of Emergency Life-saving Drugs May 2021 - Oct 2022

Supervisor: [Dr. A. B. M. Mainul Bari](#) (BUET)

Status: Published in **ELSEVIER-2022** [\[PDF\]](#)

- We identified 14 potential risk factors which can disrupt the supply chain network of the emergency life-saving drugs in Bangladesh.
- Using grey DEMATEL analysis tool, we evaluated and divided the risk factors into two groups (Cause group and Effect group). Nine risk factors were found as Cause group and rest of them were found as Effect group. Through ranking, we evaluated the most influential and resulting risk factors.
- I assisted in finding risk factors and collecting data through surveys, and conducted all the calculations to identify the Cause and Effect groups along with their ranking.
- I conducted the calculation to find the correlation between the risk factors and carried out Sensitivity Analysis to validate our results. I also took part in writing the paper draft.

Mechanical Trash Compactor

January 2021 - February 2022

Supervisor: [Dr. Kais Bin Zaman](#) (BUET)

Reference Materials: [\[Google Site\]](#)

- Explored different target customers and through survey, we identified their requirements and developed relative importance.
- We developed a QFD Model to identify the correlation between customer requirements and functional requirements. Here we prepared black box model and cluster function.
- Using Digital Logic Method, we calculated performance index for selecting material, manufacturing process and joining process. We designed a model in Solid Works and using ANSYS, we explored design analysis. Finally, we constructed the model physically.
- I conducted data collection, prepared QFD Model and took part in design analysis along with my team mates.
- I assisted in writing the scripts, designing and constructing the model.

PROFESSIONAL EXPERIENCE

IWS Process Lead

November 2022 - Present

Employer: [British American Tobacco Bangladesh](#)

- Working in the filter manufacturing department (FMD) where the team's responsibility is to maintain the filter quality and deliver it on time.
- Collaborating with stakeholders from different parts of the world and learning international standard work culture while carrying out my assigned tasks.
- Learning SAP software uses while performing Month End, MO creation, PO confirmation etc.
- Conducting trend analysis for the reduction of material wastes, quality losses, runnability losses and EHS health hazards etc.
- Implementing IWS Phase 2 Journey by using component level knowledge to provide standards and facing audits for AM Step 3 and AM Step 4 journey.

Engineering Intern

April 2022 - May 2022

Employer: [Rahimafrooz Batteries Ltd.](#)

- Worked on waste management where I identified eight types of waste and proposed countermeasures to reduce the wastes in Lean Manufacturing.
- Implemented Poka-yoke systems for safety protocol.
- Presented ergonomic uses of tools to reduce health hazards.
- Worked on line balancing to reduce bottlenecks and design the layout.

PROJECTS

Vertical Clamping Injection Molding Machine | *CATIA Project* 2022

- A 3D model of the Vertical Clamping Injection Molding Machine by using CATIA Software.

Suitcase with shelves | *Product Design Lab Project* 2021

- A project where a product is being proposed that can be used as both suitcase and rack for clothes.

Ergonomic Jack Hammer | *Ergonomic Lab Project* 2021

- Ergonomically designed a jack hammer which can provide proper gripping and can reduce health hazards.

Rome Colosseum | *Solid Works Project* 2018

- A 3D model of Rome Colosseum by using Solid Works software.

AWARDS

- Scholarship: Bangladesh University of Engineering and Technology (BUET) (Year: 2019, 2021)
- Board Merit Scholarship: Government of Bangladesh (Year: 2012, 2016)
- Honourable Mention (Top Ten) at BUET HULT Prize 2020
- Spotlight: BATB

TECHNICAL SKILLS

Programming Languages: Python, C, MATLAB

Modelling Software: AutoCAD, Solid Works, CATIA

Design Analysis Software: ANSYS

Simulation and Automation Software: Arena

Business related Software: SAP, MS Excel, MS Word, MS PowerPoint, Power App

REFERENCE

- [Dr. Abdullahil Azeem](#)
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- [Dr. A. B. M. Mainul Bari](#)
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